

Youqiu Zhan

 [ulysseszh.github.io](https://github.com/ulysseszh)

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Education and coursework

University of California, Los Angeles

2026–

Doctor of Philosophy in physics

University of California, Santa Barbara

2022–2026

Bachelor of Science in physics

(GPA: 3.93/4.0)

- Quantum many-body problem (PHYS 217A–B) A, A+
Final projects: [Quantum many-body theories about the crust of neutron stars](#), [Axion electrodynamics in topological insulators](#)
- Relativistic quantum field (PHYS 221B–C) A, audit
- Condensed matter (PHYS 223A–C) A+, A–, A
Final project: [Exact results in 2D percolation](#)
- Gauge theory (PHYS 229A–B) audit
- General relativity (PHYS 231A–C) audit, A, A
Final project: [Spinor fields in curved spacetime](#)
- String theory (PHYS 230A) audit
- Thesis project: [The broken stick problem as the continuum limit of a Bernoulli process](#)

High School Affiliated to Shanghai Jiao Tong University

2019–2022

International Baccalaureate

(Grade: 42/45)

Research experience

Bouwmeester Lab

2024–2026

- Using COMSOL to simulate phononic crystals on a thin plate to find band gaps and design ultra-high- Q mechanical resonators.
- Using COMSOL to simulate third sound modes of superfluid helium on 2D geometries.
- Using KLayout to design and produce masks used in nanofabrication of mechanical devices used in optomechanical experiments.
- Using QuTiP to simulate and visualize optomechanical processes such as stimulated adiabatic Raman passage (STIRAP).
- Using QuTiP to simulate quantum models of third sound modes of superfluid helium on 2D geometries for optomechanical systems.

UCSB honor thesis

2025–2026

Studying the continuum limit of statistical problem of the distribution of the maximum run length in a Bernoulli sequence using techniques from mathematical physics, such as the Adomian decomposition method and the grand canonical ensemble. Monte Carlo methods were used to simulate the discrete problem, in order to verify theoretical results.

Independent research

2021

Using PyTorch to implement and train self-supervised neural networks for finding the Hamiltonian of classical mechanical systems from trajectory data.

Regeneron ISEF

2020–2021

Using web technology to simulate and visualize classical Hamiltonian mechanical systems. Implementing integrators and signal processing algorithms in JavaScript.

Independent research

2020–2021

Using ABAQUS to simulate and visualize stress distribution in 3D structures fatigue life under cyclic loading conditions.

Awards

Research Excellence Award

May 28, 2026

Awarded on behalf of the UCSB physics department for demonstrating excellence in laboratory research in the field of physics.

Bachelor's Honors Thesis

May 28, 2026

Awarded on behalf of the UCSB physics department for completion of a Bachelor's Honors Thesis.

Development projects

Individual projects:

alda-rb	Ruby library and DSL for live-coding music, powered by Alda
DisplayToggle	Android app for controlling the power mode of the physical display
mechsimul-js	Hamiltonian mechanics simulator written in JavaScript
Sunniesnow	Web rhythm game inspired by Lyrica
SunProxy	Android app for setting up network packet redirection rules by VPN

Contributed projects:

AHK_X11	AutoHotkey for X11-based systems
Nixpkgs	Software package collection distributed with Nix
Paru	Ruby wrapper over Pandoc
PixiJS	2D rendering engine built on WebGL and WebGPU

Publications

Hedgepeth, Ian et al. (Aug. 2026). “Modeling quantum optomechanical stimulated Raman adiabatic passage”. en. In: *Physical Review A* 114.2, p. 023509. ISSN: 2469-9926, 2469-9934. DOI: [10.1103/mtd9-442c](https://doi.org/10.1103/mtd9-442c).

Zhan, Youqiu (Jan. 2021). “Self-supervised hamiltonian mechanics neural networks”. In: *2021 IEEE International Conference on Consumer Electronics and Computer Engineering (ICCECE)*. Guangzhou, China: IEEE, pp. 271–275. ISBN: 978-1-7281-8319-0. DOI: [10.1109/ICCECE51280.2021.9342165](https://doi.org/10.1109/ICCECE51280.2021.9342165). URL: <https://ieeexplore.ieee.org/document/9342165/>.

Zhan, Youqiu and Zhe Wu (Oct. 2020). “Reliability Analysis of Wing Fuselage Docking Structure”. In: *2020 2nd International Conference on Artificial Intelligence and Advanced Manufacture (AIAM)*. Manchester, United Kingdom: IEEE, pp. 206–215. ISBN: 978-1-7281-9986-3. DOI: [10.1109/AIAM50918.2020.00048](https://doi.org/10.1109/AIAM50918.2020.00048). URL: <https://ieeexplore.ieee.org/document/9425956/>.

Skills

- Programming languages: Ruby, JavaScript (and TypeScript), Java (and Kotlin), Python
- DevOps and software distribution: GNU/Linux, Docker, SSH, Git, Nix
- Development tech stacks: Android SDK, Node.js, Electron.js, Jekyll, Pandoc
- Scientific software: COMSOL, KLayout, BEAMER, Wolfram, Praat
- Scientific Python packages: QuTiP, NetKet, PyTorch, OpenCV
- Spoken languages: English, Mandarin Chinese